Applied Biostats Fall 2026. Practice Exam 1. Adapted from Last year's exam

Your NAME:

Answers here

Q1) **A** **B** **C**

Q2) **A** **B** **C**

Q3) **A** **B** **C** **D** **E** **F** **G**

Q4) **A** **B** **C** **D** **E**

Q5) **A** **B** **C** **D** **E** **F**

Q6) **A** **B** **C** **D** **E**

Q7) **A** **B** **C**

Q8) **A** **B** **C** **D**

Q9) **A** **B** **C** **D** **E** **F**

Q10.1) _____

Q10.2) _____

Q11) **A** **B** **C** **D** **E** **F**

Q12) **A** **B** **C** **D**

Q13) **A** **B** **C** **D**

Q14) **A** **B** **C**

Q15) _____

Q16) Draw your plots on the exam page

Q17) **A** **B** **C**

Q18) **A** **B** **C** **D**

Q19) **A** **B** **C** **D**

Q20) **A** **B** **C** **D** **E**

Q21) **A** **B** **C** **D** **E**

Q22) **A** **B** **C** **D**

Q23) **A** **B** **C** **D** **E**

Q24) **A** **B** **C** **D**

Q25) **A** **B** **C** **D**

Q26) **A** **B** **C** **D**

Q27) **A** **B** **C** **D**

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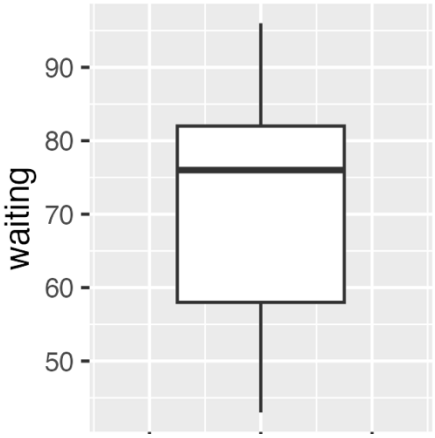
Q1. A standard rule in data visualization is to show the data (i.e. all the data points). Say I had a ton of data and didn't want to show every data point. Which would convey both a measure of *central tendency and of variability*? **Pick all correct**

A) A point for the mean, with bars showing plus or minus one standard error.
B) A point for the mean, with bars showing plus or minus one standard deviation.
C) A point for the mean, with bars showing the 95% confidence interval.

Q2. A standard rule in data visualization is to show the data (i.e. all the data points). Say I had a ton of data and didn't want to show every data point. Which would convey both the *mean of the data and the uncertainty in it?* **Pick all correct**

A) A point for the mean, with bars showing plus or minus one standard error.
B) A point for the mean, with bars showing plus or minus one standard deviation.
C) A point for the mean, with bars showing the 95% confidence interval.

Q3) Which values can you reasonably estimate from this plot (select two best)



A) Mean
B) Median
C) Mode
D) The shape of the data
E) Standard Deviation
F) Interquartile range
G) 95% confidence interval

Q4) You conduct the same study twice, once with a sample of size ten, and once with a sample of size 100. Which estimate(s) is (are) most likely to get smaller in the study with a larger sample? **select all correct**

A) Mean
B) Standard Deviation
C) Variance
D) Coefficient of variation
E) Standard error

Q5) You conduct the same study four times, once with a sample of size ten (n=10), once with n=20, once with n = 1000 & once with n = 10000. Which estimates will be most similar?

A) The means of samples with n = 10, and n =20
B) The means of sample n = 1000, and of sample n = 10000
C) The standard error of the sample with n = 1000 and n =10000
D) A & B
E) A & C
F) B &C

Q6) A student says, “The sampling distribution shows how the values in my dataset are spread.” Why are they wrong?

A) Because the sampling distribution describes variation in estimates across repeated samples, not variation within a single sample.
B) Because the sampling distribution only applies to categorical data.
C) Because the sampling distribution is calculated before collecting data.
D) Because the sampling distribution depends on the population size, not the sample size.
E) They aren’t wrong

Q7. Why is understanding the sampling distribution essential for doing inference?

A) It allows us to quantify sampling bias.
B) It allows us to quantify sampling error.
C) It allows us to quantify non-independent sampling.

Q8) Because turtle hatchlings are difficult to catch, you scatter food to attract them and measure the running speed of the first 30 individuals that reach the food. This sampling method likely introduces…

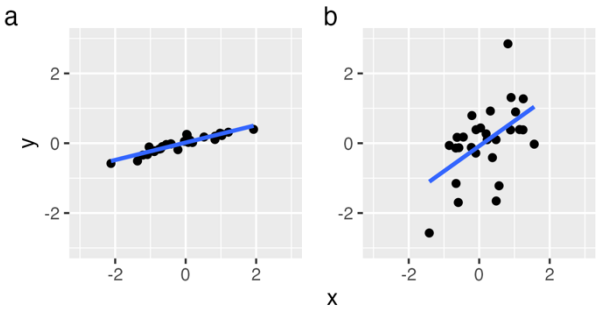
A) Sampling error, because estimates differ across samples by chance.
B) Sampling bias, because faster turtles are more likely to be measured.
C) Non-independence, because turtles from the same nest share genetics.
D) Measurement error, because timing turtles is imprecise.

Q9) A p-value of 0.01 means: (select all correct)

A) We reject the null hypothesis
B) We fail to reject the null hypothesis
C) The null is false
D) There is a 1% chance the null is correct
E) There is a 1% chance the null generated our data (or something more extreme)
F) 1% of samples from the null will be as or more extreme than our observation.

Q10.1) Plot _____ has the greater covariance.

Q10.2) Plot _____ has the greater correlation.



Q11) In statistics we usually use an alpha threshold of 0.05. What would happen if the field changed the custom to alpha = 0.01 (select all correct)?

A) We would reject more false nulls
B) We would reject more true nulls
C) We would fail to reject more false nulls
D) We would fail to reject more true nulls
E) We would have wider confidence intervals (we use 99% confidence intervals)
F) We would have narrower confidence intervals (we use 99% confidence intervals)

Q12) Suppose I'm surveying people's ice cream preferences. I have a variable in my dataset named "deliciousness", in which I had people choose from the options: "yuck", "bad", "meh", "good", "splendid", and "best I've ever had". What kind of variable is this?

A) Numeric B) Binary C) Ordinal D) Continuous

Q13) "jittering" is a great way to prevent "overplotting" but it can go wrong. What should you watch out for (select all correct):

A) Be sure not to fool readers into thinking x is continuous
B) Make sure not to trick readers into wrongly thinking the jitter around y-values are true variation.
C) Make sure that readers can differentiate between categories on x.
D) Chill, jitter plots cannot be messed up.

The penguins data set in R describes morphology of hundreds of penguins from three species and three islands. Here I am looking at the difference in bill length (Male - Female) of Adelie penguins on Dream island. You will need these summaries for the next few questions:

- Mean male: 40.1
- Estimated difference: 3.23
- 95% CI for difference: 2.08 to 4.38
- p-value: 0.00000122
- Cohen's d : 1.64

Q14) The correct interpretation of the confidence interval is (select all correct)

A) There is a 95% chance the true mean sex difference is between 2.08 and 4.38
B) If this study were repeated many times, 95% of those confidence intervals would include the true mean sex difference bill length. We don't know what this is, but from our study a reasonable guess is between 2.08 and 4.38
C) If I randomly pair male and female Adelie penguins from Dream island, 95% of pairs will have a difference in bill length within 2.08 and 4.38.

Q15) Find the pooled standard deviation.

Q16) From the set-up above draw two plots (as either histograms or density plots). Provide enough detail and labeling such that a reader could roughly infer these values from the plots (assume roughly equal variance between sexes).

Q17) I found the p-value by permutation. This means

A) I resampled the data with replacement, calculated the mean difference, repeated this many times, and found the proportion of these that were less than zero
B) I swapped labels of males and females at random, calculated the sex difference in bill length, repeated this many times, and found the proportion of replicates with an absolute value greater than 3.23.
C) I sampled from the population many times and found the proportion of sample mean differences with an absolute value greater than 3.23.

Q18) I am curious to compare the sex differences in bill length to the sex difference in body mass. Which comparison is most insightful?

A) Comparing Cohen's d for sex difference in bill length to Cohen's d for sex difference in body mass.
B) Comparing the p-value for sex difference in bill length to the p-value for sex difference in body mass.
C) Comparing the sex difference in the conditional means in bill length to the sex difference in the conditional means in body mass.
D) Comparing the correlation between bill length and body mass between the sexes.

Q19) In attempting to find these calculations I typed the following and got the accompanying error

```
> penguins|> filter(species == "Adelie")
```

```
Error: object 'species' not found
```

Show Traceback
Rerun with Debug

To follow up, I typed `> View(penguins)`

And got

	species	island	bill_len	bill_dep	flipper_len
1	Adelie	Torgersen	39.1	18.7	181
2	Adelie	Torgersen	39.5	17.4	186

What went wrong?

A) Adelie should not be in quotes
B) species should be capitalized
C) filter() should be on a different line than penguins
D) I didn't tell R which filter function to use.

Q20) A histogram of pollinator visits per plant shows that most plants receive zero visits and a few receive many (strongly right-skewed). Which are true? (select all correct)

A) The mean will be greater than the median.
B) The median will be greater than the mean
C) The median better describes a "typical" plant than the mean does
D) Because the data are skewed, we cannot calculate a standard deviation
E) Skewed data always have a larger variance than symmetric data

Q21) You give fertilizer to one greenhouse bench and none to another, then measure 50 plants on each bench. What is the problem?

A) A sample of 50 plants per treatment is too small.
B) Data are not independent.
C) There is sampling bias
D) The researcher should have been blinded.
E) There's no problem.

Q22) In plain words, the standard deviation of a sample roughly conveys: (pick the best answer)

A) The difference between the largest and smallest values.
B) How far a typical data point is from the mean.
C) How much the sample mean would change if we sampled again.
D) The proportion of data points that fall far from the mean

Q23) In plain words, the standard error of a sample mean roughly conveys: (pick the best answer)

A) How far a typical data point is from the mean.
B) The amount of error introduced by sampling bias.
C) How far a sample mean typically is from the true population mean.
D) The probability that our estimate is wrong.
E) Mistakes made while measuring or recording the data.

Q24) The key difference between an experimental and an observational study is that, in an experiment: (pick the best answer)

A) The study takes place in a lab rather than in the field.
B) The researcher assigns the values of the explanatory variable (ideally at random).
C) The sample size is larger.
D) There is a comparison group.

Q25) I measured hemoglobin concentration in people from four populations (USA, Andes, Ethiopia and Tibet). There are two ways I could have stored these data. Which is tidy?

Table A

hemoglobin	population
15.6	Tibet
15.8	USA
19.1	Andes
15.9	Ethiopia
15.2	USA
15.2	Tibet
18.6	Andes
16.3	Ethiopia

Table B

USA	Andes	Ethiopia	Tibet
15.8	19.1	15.9	15.6
15.2	18.6	16.3	15.2

A) Table A
B) Table B
C) Both are tidy
D) Neither is tidy

Q26) Table B has a presentation problem. What is it? (pick the best answer)

A) The numbers are rounded.
B) It gives the appearance of row as a meaningful but here a row has no real meaning, .
C) It has too many columns.
D) It uses population names instead of numbers.

Q27) I wrote the script below while working through my analysis, and everything worked on my computer. I saved it, closed R, and reopened it. Now, when I run the script from top to bottom, I get the error: object 'gc_rils' not found. Why? (pick the best answer)

```
library(dplyr)
library(readr)

gc_summary <- gc_rils |>
  group_by(petal_color) |>
  summarise(mean_hybrid = mean(prop_hybrid, na.rm = TRUE))

gc_rils <- read_csv("clarkia_rils.csv")|>
                dplyr::filter(location == "GC")
```

A) The data file must have changed after I closed R.
B) The script uses gc_rils before the line that creates it. It only worked before because gc_rils was already in my R session from running lines out of order.
C) I need to reinstall dplyr every time I open R.
D) R scripts cannot be saved and rerun later.